

Attachment B – Technical Requirements

This worksheet is intended to provide an overview of both the operational and data requirements for the anticipated UAS platform. Vendors should reference these desired capabilities when providing Rough Order of Magnitude (ROM) pricing in Attachment C: ROM Pricing Template.

Section 1 – Operations

Minimum Operational Requirements

The following is a comprehensive list of the minimum operational requirements:

FRA UAS Program – Minimum Operational Requirements		
Category	Requirement	Details / Notes
Platform Capabilities	VTOL	Vertical takeoff and landing for constrained environments such as rail corridors and grade crossings.
	Autonomous Flight	Fully autonomous flight with advanced obstacle avoidance and path planning; manual override for pilot intervention.
	Ruggedized	Built for harsh weather and railroad field conditions.
	Max Payload	Capable of carrying cameras, LiDAR, and other sensors.
	Parachute	Integrated emergency recovery system for safe descent in case of critical failure.
Flight Performance	Endurance	Minimum 30–60 minutes per battery cycle, depending on payload.
	Operational Range	Up to 10 km from control station or BVLOS
	Wind Tolerance	Suitable for open corridor and urban operations.
	Batteries	Rapid swap capability; extra batteries required for extended operations.
	Propellers	Spare propellers available for field replacement.
Payload / Sensors	Optical Camera	High-resolution still imagery and 4K+ video recording.
	LiDAR / 3D Mapping	For photogrammetry and 3D model creation.
	Thermal Imaging	For infrastructure inspections and emergency response.
	Stabilized Gimbal	Automated imaging controls for repeatable inspection flights.
	Additional Payloads	Optional GNSS or other sensors for high-accuracy mapping.
	Ground Control Points	Support for at least 6 GCPs per mission for accurate geospatial mapping.

Data Management	Storage	Onboard encrypted storage; automated offload after landing.
	Live Feed	Real-time video streaming for situational awareness.
	Photogrammetry	Orthomosaics, 3D models, and integration with mapping software.
	Integration	Compatible with FRA flight tracking and data management workflows.
Safety / Compliance	FAA Regulations	Part 107 or other applicable authorizations.
	USG Directives	NDAA Sec. 848 (FY2020), Sec. 817 (FY2023), and American Security Drone Act compliant.
	Redundant Systems	Dual IMUs, GPS, obstacle avoidance sensors for failsafe operation.
	Remote Pilot Interface	Geofencing, safety alerts, and full control override.
Operational Support	Transport	Compatible with secure truck vaults for UAS and equipment storage.
	Charging	Portable field charging systems for multi-mission days.
	Workflow Integration	Supports automated mapping and live-feed review.
	Software Updates	Platform supports ongoing development and FRA-specific requirements.
Environmental / Field Considerations	Weather & Terrain	Operational in rain, dust, moderate temperatures; designed for rapid deployment.
	Field Deployment	Adapted for rail-adjacent, emergency response, and difficult-to-access sites.

All operations must comply with FAA regulations, U.S. government directives, and FRA policies. Full compliance and security regulations are provided in Attachment A.

Operational Use Cases

The FRA envisions the following scenarios where a UAS platform can enhance operational efficiency. Vendors should ensure adaptability with the following for the proposed delivery model and coinciding operational capabilities.

Scenario	Description	Frequency
Grade crossing inspection	Capture aerial imagery of crossings to evaluate visibility and obstruction conditions	Weekly
ENS Inspection	Ensure compliance of ENS Signage	Weekly

Scenario	Description	Frequency
Derailement Investigation	Assist accident branch in conducting photogrammetry analysis of derailments to locate cause.	As Needed
Hazard Material Inspection	Conduct inspection of hazard material incidents for releases and provide live feeds to assist with the containment	As Needed
Incident response	Post-incident aerial documentation	As needed
Infrastructure Assessment	Assist OST and FRA conducting infrastructure assessments	As Needed
Site Distance Reviews	The purpose is to document sight distances at highway-rail grade crossings after a grade crossing incident or at the request of the accident branch. A determination of the site distance will be determined whether vegetation or other obstructions have interfered with motorist visibility of highway-rail grade crossing warning devices and or on-coming train. Whether on railroad property or off railroad property.	As need
Quiet Zone Inspection	Quiet Zone is a total physical grade crossing and all its attributes including roadway elements, signage (excluding ENSS), SSM's, ASM's. inspected for accuracy required by 49 CFR Pt 222.	Weekly
Trespassing Detection	Conduct arial inspections of ROW to locate high areas of railroad trespassing	Weekly

Operational locations include, but are not limited to:

- Public right-of-way
- Railroad property
- Remote rural crossings
- Urban crossings
- Training Environments (e.g., Transportation Technology Center)
- Incident response locations

UAS operations may support inspections and assessments related to railroad infrastructure, grade crossings, quiet zones, Emergency Notification Systems (ENS), trespassing locations, and other safety-related conditions. Operations may also occur in support of incident response situations, including hazardous material incidents, derailments, hurricanes, severe weather events, or other emergencies where aerial assessment of railroad infrastructure and surrounding conditions is necessary.

Operational Assumptions

Vendors may assume the following indicative operational ranges when providing pricing estimates:

- Low Utilization: 10–25 missions per week
- Medium Utilization: 25–75 missions per week
- High Utilization: 75–150 missions per week
- Estimated derailment response events: approximately 25 per year.

Section 2 – Data

Minimum Requirements

The following reflect the data intended to be collected through UAS capabilities:

- Standard aerial imagery
- High-resolution photography
- Photogrammetry
- Thermal imagery
- Video capture

UAS operations will support the collection of various types of data and imagery to assist FRA inspectors in conducting safety inspections and investigations. This may include high-resolution aerial imagery, live video feeds for situational awareness, and photogrammetry datasets used to generate orthomosaics, 3D models, and other geospatial products. These data products will support activities such as site distance analysis, measurements of roadway and railroad infrastructure, documentation of accident scenes, and assessment of conditions related to regulatory compliance. The imagery and data collected will assist inspectors in documenting findings, conducting measurements, and maintaining accurate records of inspection activities.

Data Usage

The data and imagery collected through UAS operations will be used to support FRA inspectors in evaluating railroad infrastructure, documenting inspection findings, and assessing compliance with applicable federal safety regulations. The information may be used to conduct measurements, perform site distance analysis at grade crossings, document accident or incident scenes, and identify safety concerns such as trespassing locations or infrastructure deficiencies. The collected data will also support the creation of inspection records, reports, and geospatial products that assist FRA in making informed regulatory and safety oversight decisions.

Data Assumptions

All imagery, metadata, and derived geospatial products generated through UAS operations in support of FRA missions shall remain the property of the Federal Government.

Solutions proposed under this RFI should support integration with FRA enterprise data systems and architectures, including geospatial platforms, inspection systems, records management systems, and analytics environments

Solutions that rely exclusively on proprietary vendor-hosted data environments without the ability to export and integrate data into government systems may limit the operational utility of the platform.

Vendors should also reference the following data and cybersecurity assumptions:

- Secure data transmission (LTE/SATCOM as applicable)
- FISMA-aligned architecture
- Continuous monitoring capability
- Data retention period: 3–5 years
- Integration capability with government systems

